

OS Lab experiments

Program1:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ cat program1.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    printf("PID = %d\n", getpid());
    return 0;
}

nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program1
PID = 553
```

Program2:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ cat program2.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    printf("Hello\n");
    return 0;
}

nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program2
Hello
Hello
```

Program3:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ cat program3.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
        printf("I am Child\n");
    else
        printf("I am Parent\n");

    return 0;
}
nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program3
I am Parent
I am Child
```

Program4:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ cat program4.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program4
Parent PID = 688
Child PID = 689
Child PID = 689
Parent PID = 688
```

Program5:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ cat program5.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child Executing\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Executing\n");
    }
    return 0;
}

nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program5
Child Executing
Parent Executing
```

Program6:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ cat program6.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();
    fork();
    printf("PID = %d\n", getpid());
    return 0;
}

nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program6
PID = 730
PID = 732
PID = 731
PID = 733
```

Program7:

```

nishanth@LAPTOP-6Q3ON07H:~/fork$ cat program7.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child executing ls...\n");
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else
    {
        printf("Parent Waiting...\n");
    }
    return 0;
}
nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program7
Parent Waiting...
Child executing ls...
nishanth@LAPTOP-6Q3ON07H:~/fork$ total 160
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 29 03:17 program1
-rw-r--r-- 1 nishanth nishanth  99 Jul 29 03:24 program1.c
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 29 03:25 program2
-rw-r--r-- 1 nishanth nishanth  104 Jul 29 03:25 program2.c
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 29 03:34 program3
-rw-r--r-- 1 nishanth nishanth  197 Jul 29 03:33 program3.c
-rwxr-xr-x 1 nishanth nishanth 16088 Jul 29 03:36 program4
-rw-r--r-- 1 nishanth nishanth  338 Jul 29 03:36 program4.c
-rwxr-xr-x 1 nishanth nishanth 16040 Jul 29 03:40 program5
-rw-r--r-- 1 nishanth nishanth  215 Jul 29 03:39 program5.c
-rwxr-xr-x 1 nishanth nishanth 16040 Jul 29 03:43 program6
-rw-r--r-- 1 nishanth nishanth  117 Jul 29 03:42 program6.c
-rwxr-xr-x 1 nishanth nishanth 16040 Jul 29 03:53 program7
-rw-r--r-- 1 nishanth nishanth  220 Jul 29 03:53 program7.c
-rwxr-xr-x 1 nishanth nishanth 16040 Jul 29 03:58 program8
-rw-r--r-- 1 nishanth nishanth  133 Jul 29 03:55 program8.c

```

Program8:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ nano program9.c
nishanth@LAPTOP-6Q3ON07H:~/fork$ gcc program9.c -o program9
nishanth@LAPTOP-6Q3ON07H:~/fork$ ./program9
Enter Linux Command: ls

Parent Process
Parent PID : 950
Child PID  : 951

Child Process
Child PID : 951
Parent PID: 950
program1 program1.c program2 program2.c program3 program3.c program4 program4.c program5 program5.c
```

```
program6 program6.c program7 program7.c program8 program8.c program9 program9.c
```

```
Child Process Completed
```

History:

```
nishanth@LAPTOP-6Q3ON07H:~/fork$ history | sed -n '33,71p'
```

```
33 mkdir fork
34 cd fork
35 nano program1.c
36 gcc program1.c -o program1
37 nano program1.c
38 gcc program1.c -o program1
39 ./program1
40 nano program2
41 gcc program1.c -o program2
42 ./program2
43 ./program1
44 nano program1.c
45 ls
46 rm program2
47 ls
48 nano program2.c
49 gcc program2.c -o program2
50 ./program2
51 ls
52 nano program3.c
53 gcc program3.c -o program3
54 ./program3
55 nano program4.c
56 gcc program4.c -o program4
57 ./program4
58 nano program5.c
59 gcc program5.c -o program5
60 ./program5
61 nano program6.c
62 gcc program6 -o program6
63 gcc program6.c -o program6
64 ./program6
65 ./program4
66 nano program7.c
67 gcc program7.c -o program7
68 ./program7
69 nano program8.c
70 gcc program8.c -o program8
71 ./program8
```

```
93 nano program9.c
94 gcc program9.c -o program9
95 ./program9
96 history
```

